June 11, 2026

Editorial Board
Proceedings of the National Academy of Sciences

Dear Editors,

I am pleased to submit the enclosed manuscript, "Training-Free Generation of Protein Sequences from Small Family Alignments via Stochastic Attention," for consideration as a Research Article in PNAS.

Most protein families are too small to train a deep generative model: the median Pfam seed alignment contains only 22 sequences, and nearly half of all families have fewer than 20. This paper shows that a single attention operation over a family alignment defines an energy landscape whose Boltzmann distribution can be sampled via Langevin dynamics, producing novel protein sequences with no training, no external data, and no GPU resources. The resulting method, stochastic attention, generates sequences that preserve family-level amino acid statistics and are predicted by two independent structure predictors (ESMFold and AlphaFold2) to adopt the correct three-dimensional fold. We validate the approach across eight diverse Pfam families, establish an empirical scaling law that enables fully automatic operation from a seed alignment, and confirm biological plausibility through ESM2 pseudo-perplexity scoring and deep mutational scanning cross-referencing.

We believe this work will be of broad interest to the PNAS readership for several reasons. First, it addresses a practical gap in protein engineering: the majority of protein families fall below the data threshold of current deep generative models, yet these families are precisely where generative tools are most needed. Second, the theoretical connection between attention mechanisms and Boltzmann sampling suggests a general principle that extends beyond proteins to any domain where examples are scarce but structured. Third, the method's simplicity and accessibility (it runs in seconds on a laptop) lowers the barrier to protein sequence generation for experimentalists who lack access to large-scale computing infrastructure.

To assist with editor assignment, I suggest the following members of the PNAS Editorial Board, all of whom oversee Biophysics and Computational Biology and are well placed to handle this manuscript:

- Professor J. Andrew McCammon (University of California, San Diego), an expert in computational biophysics and the statistical mechanics of biomolecules.
- Professor William F. DeGrado (University of California, San Francisco), an expert in de novo protein design.
- Professor Susan Marqusee (University of California, Berkeley), an expert in protein folding thermodynamics and energy landscapes.
- Professor Axel T. Brunger (Stanford University), an expert in computational methods for structural biology.

I also suggest the following members of the National Academy of Sciences, each an expert in the energy-landscape and sequence-statistics methods on which this work is based:

- Professor Peter G. Wolynes (Rice University), whose energy-landscape theory of proteins is directly relevant to the Boltzmann-sampling framework used here.
- Professor Arup K. Chakraborty (Massachusetts Institute of Technology), an expert in the statistical physics of protein sequences and coevolution.
- Professor Michael Levitt (Stanford University), an expert in computational structural biology.

I also suggest the following qualified reviewers, none of whom has co-authored with me or has a conflict of interest with this work:

- Professor Debora S. Marks (Harvard Medical School), an expert in deep generative models of protein sequences learned from family alignments and in relating sequence variation to function.
- Professor Birte Höcker (University of Bayreuth), an experimental protein engineer whose group designs and characterizes proteins using evolution- and family-informed approaches, well placed to evaluate the practical utility of the method for protein engineering.
- Professor Possu Huang (Stanford University), an expert in de novo protein design and engineering whose group experimentally characterizes designed proteins, able to assess the method's value for generating foldable, functional sequences.
- Professor Sarel J. Fleishman (Weizmann Institute of Science), a developer of computational protein design methods validated extensively in the wet lab, well placed to judge whether the generated sequences are useful starting points for experimental protein engineering.
- Dr. Dmitry Krotov (MIT-IBM Watson AI Lab), a developer of modern Hopfield networks and dense associative memory, the energy landscape on which our sampler is based.
- Professor Sergey Ovchinnikov (Massachusetts Institute of Technology), an expert in protein coevolution, structure prediction, and sequence design, well placed to assess our structural and covariation analyses.

This manuscript has not been published and is not under consideration elsewhere. A preprint is available on arXiv (2603.14717).

Sincerely,

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